

CO2 - AT1 - Test.

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Course:- Computer vision

Code:- ITA0502

Slot:- C

21) Given:-

a) Gray-scale image is enhance using a -ve transformation. At an Image by brightness of its Pixels. A -ve highlights areas as dark and darkness of light.

Formula:-

$$S = 255 - x.$$

b) Original Pixel:-

50
100
150
200

Grey scale Pixel:-

205
155
105
55

∴ The Image becomes a -ve Image. bright regions become dark and dark regions become bright.

Histogram equalization:-

* Histogram equalization improves Image contrast by increasing Intensity at an Image.

Cumulative distribution function (CDF)

Formula:

Intensity

Frequency

CDF

0

1

2

3

4

6

10

6

4

10

20

26

Total Pixel = 26

Grand Total Intensity value

$$\text{New Intensity} = \left(\frac{\text{CDF}}{26} \right) * 255$$

Intensity

CDF

New Intensity

0

1

2

3

Pixel

4

10

16

26

values

0

1

2

3

in the image the image is cleared with contrast

Global

Smoothing

to convert back it

low

high

to convert back it

Given,

Gaussian kernel $\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$ Image region $\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$

$$= (10 + 20 + 10 + 20 + 40 + 20 + 10 + 20 + 10) / 16$$

$$= \frac{360}{16} = 22.5 \approx 23$$

∴ The Center Pixel becomes 23.

Edge detection using Sobel operation:

→ It finds the boundaries of objects by detecting sudden changes in pixel intensity.

Given,

Sobel mask $\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$ Image $\begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$

$$\Rightarrow (-10) + (-20) + (-10) + 0 + 0 + 0 + 30 + 60 + 30$$

$$\Rightarrow 120 - 40$$

$$= 80$$

→ The gradient value is 80, indicating a strong horizontal edge.

Q) Contrast Stretching

Contrast stretching increases the difference between dark and bright pixels making the image easier to see.

Formula:

$$S = \frac{(x - 50) \times 255}{150 - 50}$$

Original values

Calculation

New values:

50

0

0

100

128

128

150

255

255

* The linear contrast stretching to map the range [50, 150] to [0, 255], and with newly transformed values.